23CCE311

SOFTWARE DEFINED RADIO ARCHITECTURE

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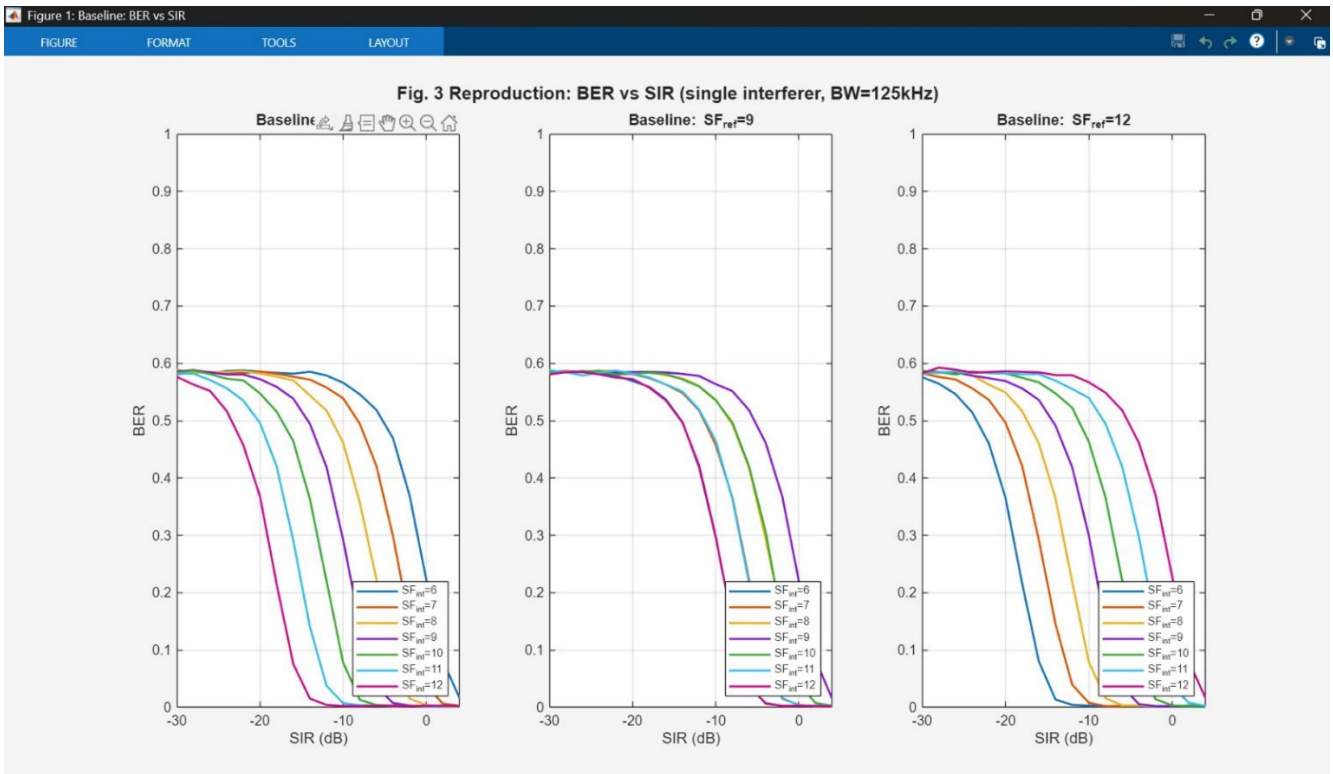
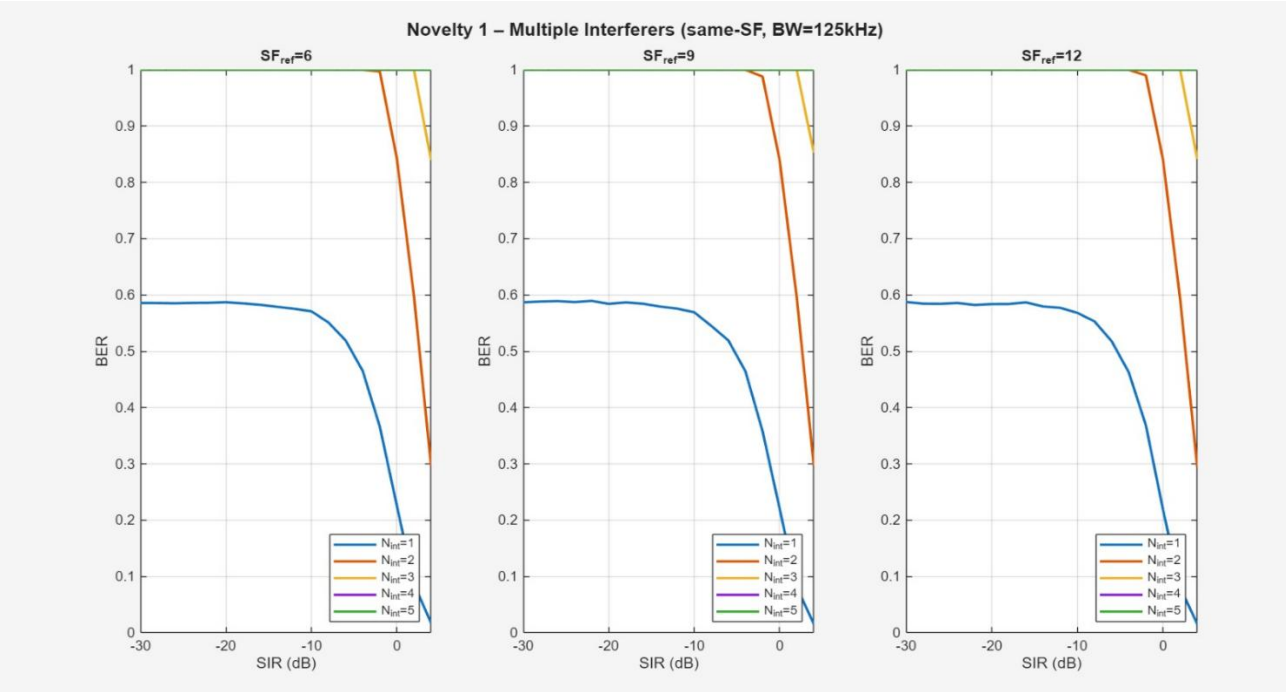
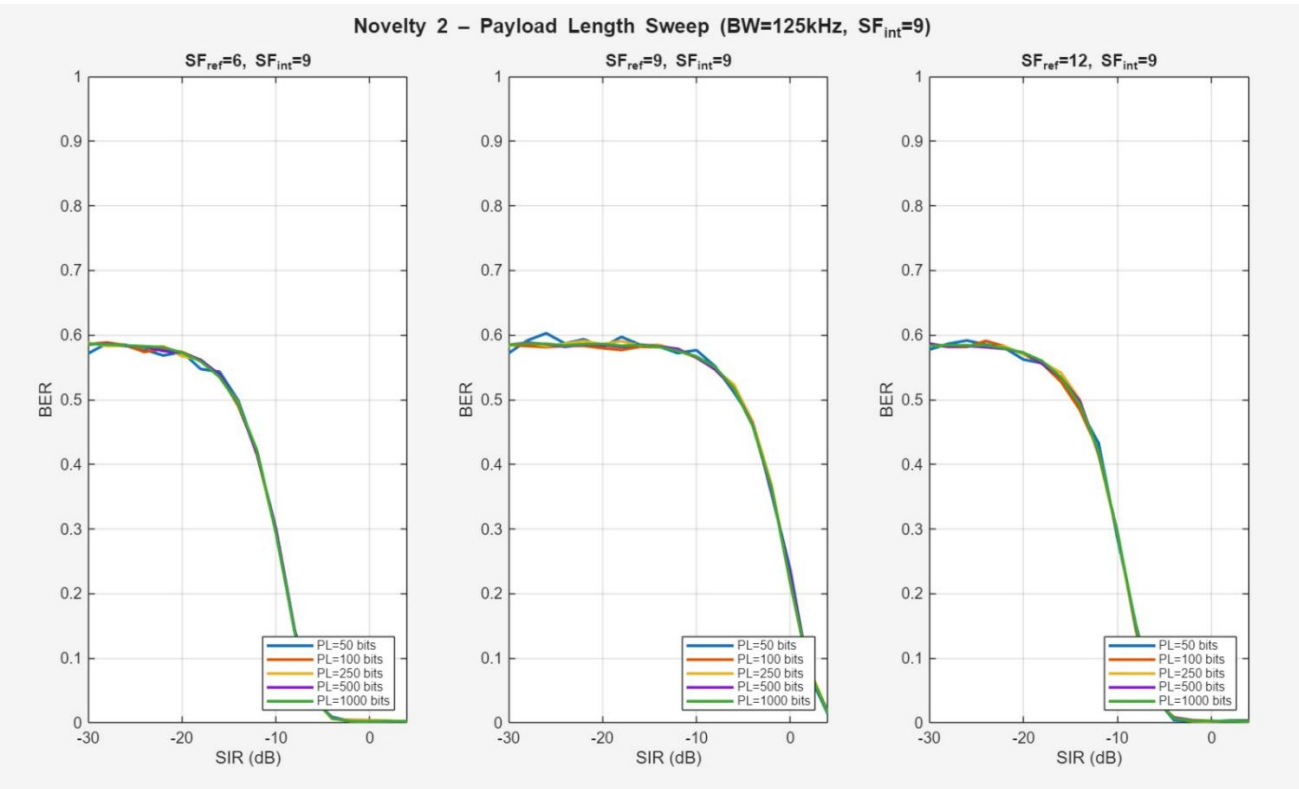
OUTPUTS:



BER decreases significantly with increasing SIR for all spreading factor combinations, indicating improved decoding reliability as interference reduces. Higher reference spreading factors demonstrate better robustness against interference, confirming the quasi-orthogonality behaviour of LoRa signals in single-interferer scenarios.

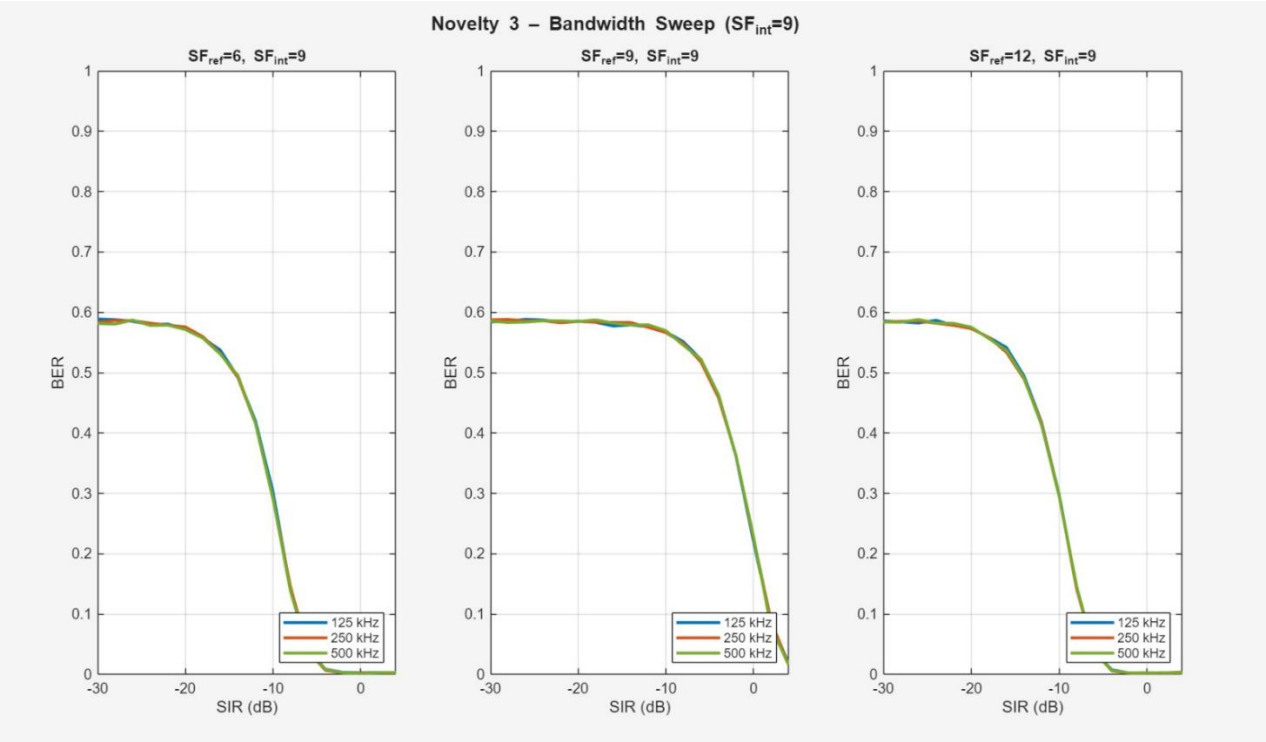


The BER increases significantly as the number of simultaneous interferers increases, even at higher SIR values, indicating severe performance degradation in dense network scenarios. This result highlights that aggregate interference from multiple same-SF transmissions reduces the capture capability and limits reliable LoRa communication.

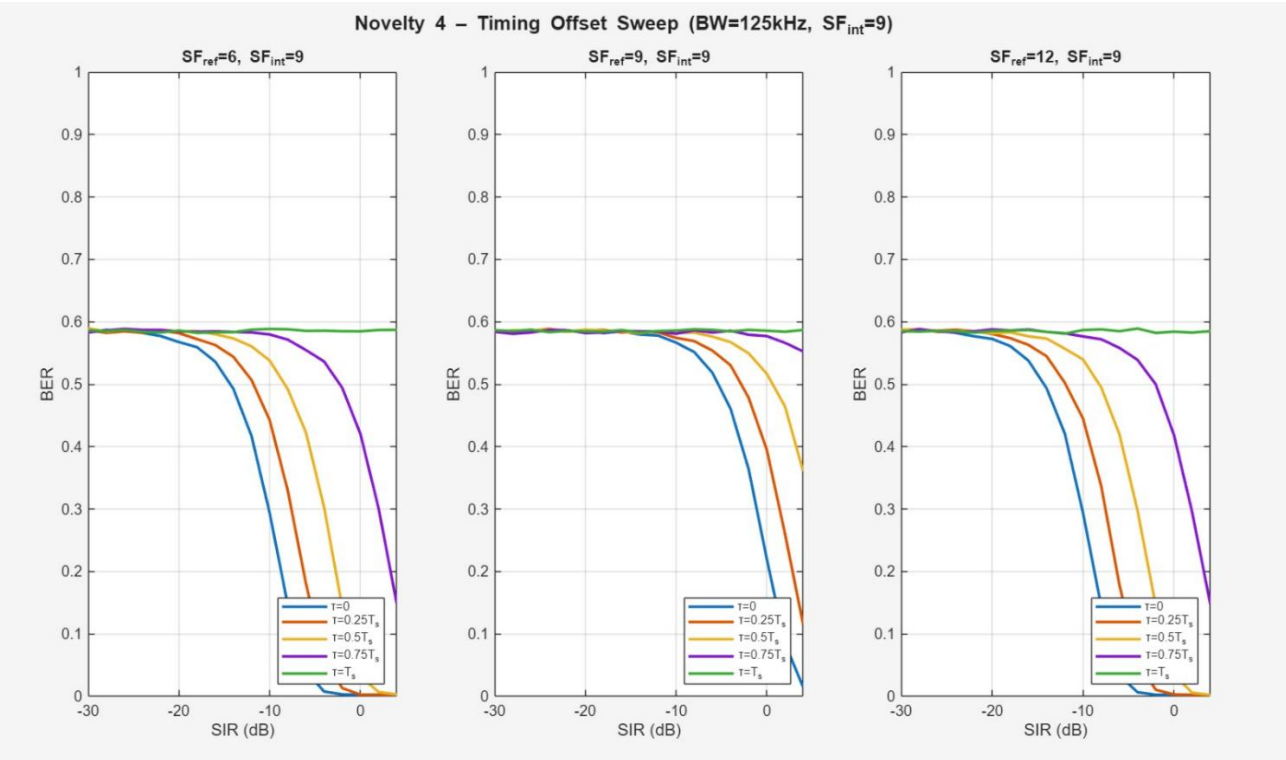


The BER performance shows only minor variation with changes in payload length, indicating that interference impact dominates over packet size in link-level decoding. However, slightly higher BER

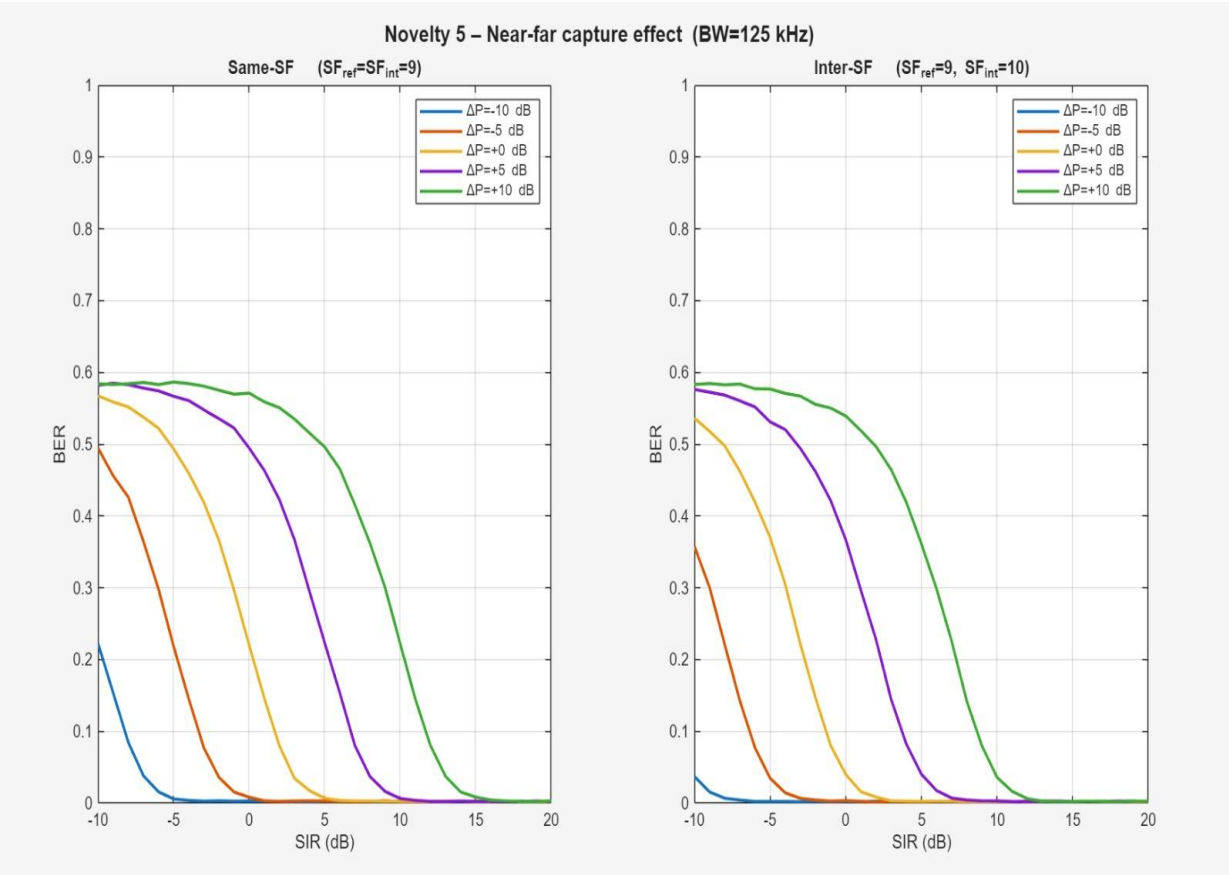
trends for larger payloads suggest increased vulnerability due to longer time-on-air and higher collision exposure in LoRa transmissions.



The BER performance shows only marginal variation across different bandwidth values, indicating that interference conditions dominate link reliability in LoRa systems. However, slight improvements at higher bandwidths suggest reduced symbol duration and lower collision probability, highlighting a trade-off between sensitivity and transmission robustness.



The BER increases noticeably as the timing offset between the reference and interfering signals increases, showing that synchronization errors significantly degrade LoRa decoding performance. Large timing misalignments reduce the correlation peak during dechirping, leading to higher symbol detection errors and reduced link reliability.



The results show that BER strongly depends on the power difference between the reference and interfering signals, demonstrating the capture effect in LoRa communication. When the reference signal is stronger (positive ΔP), reliable decoding is achieved even at low SIR, while a stronger interferer significantly degrades performance in both same-SF and inter-SF scenarios.